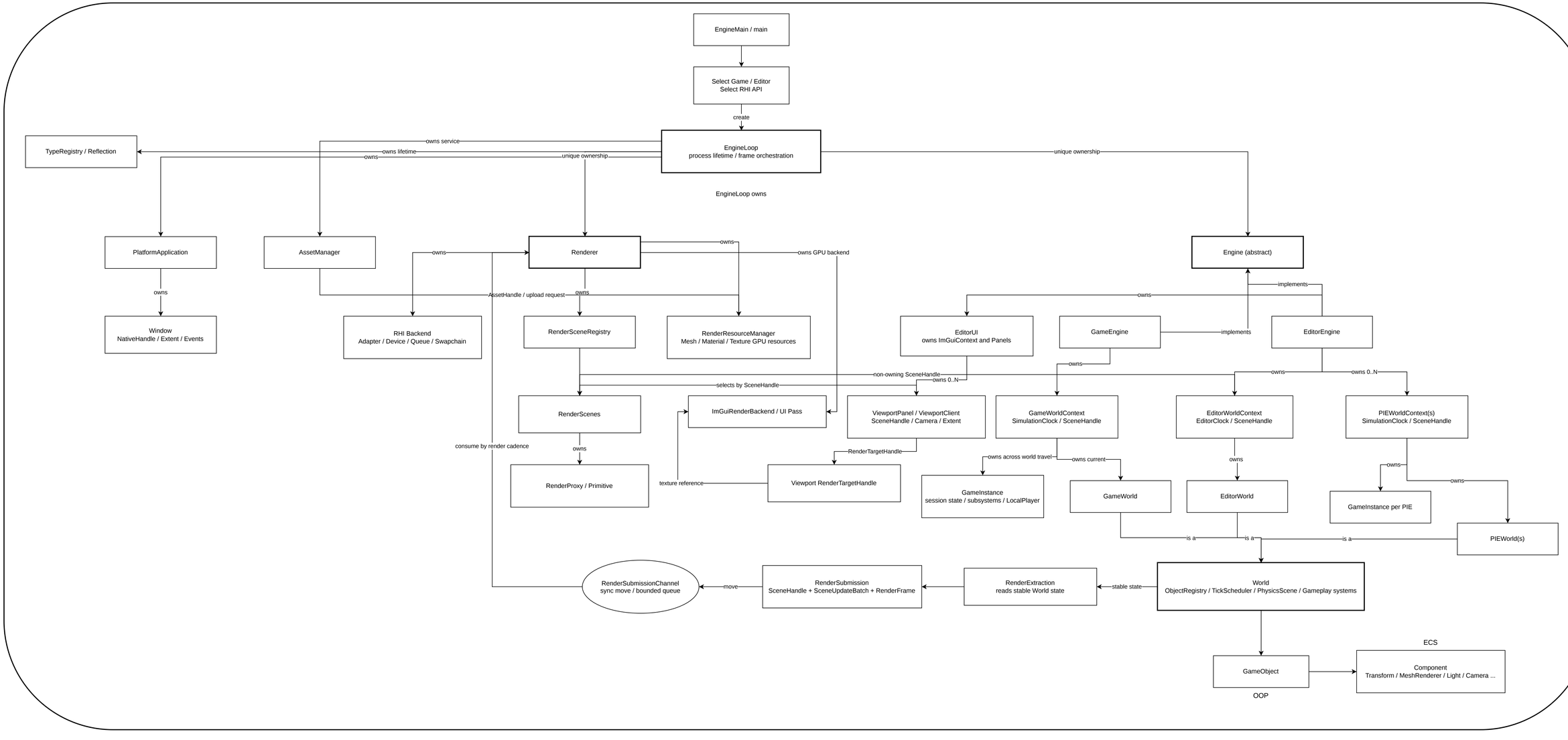
EngineMain / main
Select Game / Editor
Select RHI API
create
EngineLoop
process lifetime / frame orchestration
TypeRegistry / Reflection
owns service
owns lifetime
owns
unique ownership
unique ownership
EngineLoop owns
PlatformApplication
AssetManager
Renderer
Engine (abstract)
owns
owns
owns
owns GPU backend
owns
AssetHandle / upload request
implements
owns
Window
NativeHandle / Extent / Events
RHI Backend
Adapter / Device / Queue / Swapchain
RenderSceneRegistry
RenderResourceManager
Mesh / Material / Texture GPU resources
EditorUI
owns ImGuiContext and Panels
GameEngine
implements
EditorEngine
non-owning SceneHandle
owns
owns
owns 0..N
selects by SceneHandle
owns 0..N
RenderScenes
ImGuiRenderBackend / UI Pass
ViewportPanel / ViewportClient
SceneHandle / Camera / Extent
GameWorldContext
SimulationClock / SceneHandle
EditorWorldContext
EditorClock / SceneHandle
PIEWorldContext(s)
SimulationClock / SceneHandle
consume by render cadence
owns
RenderTargetHandle
owns across world travel
owns current
owns
owns
owns
RenderProxy / Primitive
texture reference
Viewport RenderTargetHandle
GameInstance
session state / subsystems / LocalPlayer
GameWorld
EditorWorld
GameInstance per PIE
is a
is a
is a
PIEWorld(s)
RenderSubmissionChannel
sync move / bounded queue
move
RenderSubmission
SceneHandle + SceneUpdateBatch + RenderFrame
RenderExtraction
reads stable World state
stable state
World
ObjectRegistry / TickScheduler / PhysicsScene / Gameplay systems
ECS
GameObject
Component
Transform / MeshRenderer / Light / Camera ...
OOP